

| Question |     |       | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks available |          |          |           |          |          |
|----------|-----|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |     |       |                                                                                                                                                                                                                                                                                                                                                                                                                                           | AO1             | AO2      | AO3      | Total     | Maths    | Prac     |
| 3        | (a) |       | The (vector) sum of the momenta of bodies in a system stays constant (even if forces act between the bodies) accept overall momentum remains constant [1]<br>provided there is no external / resultant force [1]<br><b>Accept:</b> Total momentum of a system (or bodies) before a collision (or explosion) = total momentum after collision (explosion)... [1]<br>....provided no external / resultant forces act [1]                    | 2               |          |          | 2         |          |          |
|          | (b) | (i)   | $600 \times 1.8 \times 10^3 = (500 \times v) + (100 \times 2 \times 10^3)$ [1]<br>$v = [+]$ 1 760 [m s <sup>-1</sup> ] [1]                                                                                                                                                                                                                                                                                                                |                 | 2        |          | 2         | 2        |          |
|          |     | (ii)  | Use of $\frac{1}{2}mv^2$ [1]<br>Before separation $E_k = 9.72 \times 10^8$ [J] [1]<br>After separation $E_k = 9.74 \times 10^8$ [J] <b>ecf</b> [1]                                                                                                                                                                                                                                                                                        | 1               | 1<br>1   |          | 3         | 2        |          |
|          |     | (iii) | Chemical energy <b>or</b> thermal energy <b>or</b> internal energy <b>or</b> work from explosion [transferred to $E_k$ ]                                                                                                                                                                                                                                                                                                                  | 1               |          |          | 1         |          |          |
|          | (c) |       | Substitution into $Ft = mv - mu$ for probe i.e.<br>$F \times 0.002 = 100 (2 \times 10^3 - 1.8 \times 10^3)$ [1]<br><b>Alternative for 1<sup>st</sup> mark:</b><br>Use of $F = ma$ to calculate $a = \frac{(v-u)}{t} = 1 \times 10^5$ m s <sup>-2</sup><br><b>Alternative for 1<sup>st</sup> mark:</b><br>$F \times 0.002 = 500 (1.76 \times 10^3$ <b>ecf</b> $- 1.8 \times 10^3)$ [1]<br>$F = [-]$ $1 \times 10^7$ N <b>unit mark</b> [1] |                 | 2        |          | 2         | 2        |          |
|          |     |       | <b>Question 3 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>4</b>        | <b>6</b> | <b>0</b> | <b>10</b> | <b>6</b> | <b>0</b> |